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Abstract

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1. Introduction

2. Randomized Network-guided Hierarchical Active Learning

We propose *Randomized Network-guided Hierarchical Active Learning* (RN-HAL), a batch deep active learning method that couples a randomized-network representation with hierarchical selection to improve diversity, coverage, and label efficiency. At active learning round t , let $\mathcal{D}_L^{(t)}$ and $\mathcal{D}_U^{(t)}$ denote the labeled and unlabeled sets, respectively. A predictive classifier f_{θ_t} is trained on $\mathcal{D}_L^{(t)}$, while the RNHAL query mechanism g_t^{RNHAL} selects a batch of unlabeled samples for annotation:

$$\mathcal{Q}_t = \Gamma_{n_{\text{query}}} \left(\mathcal{D}_U^{(t)}; g_t^{\text{RNHAL}} \right). \quad (1)$$

After annotation, the active learning sets are updated as

$$\mathcal{D}_L^{(t+1)} = \mathcal{D}_L^{(t)} \cup \mathcal{Q}_t, \quad \mathcal{D}_U^{(t+1)} = \mathcal{D}_U^{(t)} \setminus \mathcal{Q}_t. \quad (2)$$

RNHAL is motivated by a limitation of conventional acquisition functions, namely their tendency to over-select redundant samples from dense regions of the unlabeled pool when decisions rely only on predictive uncertainty or

flat feature-space criteria. To address this issue, RNHAL operates in two stages. First, each unlabeled image $x \in \mathcal{D}_U^{(t)}$ is mapped by a representation function $\Phi(\cdot)$ into an embedding

$$\mathbf{z}_x = \Phi(x), \quad (3)$$

where $\mathbf{z}_x \in \mathbb{R}^d$. In this work, $\Phi(\cdot)$ is instantiated through a spatio-spectral randomized-network formulation, so that the resulting representation captures both local spatial structure and cross-channel spectral relationships. Second, the embeddings of the current unlabeled pool are organized through hierarchical clustering, and samples are selected across the induced hierarchy. This design decouples predictive learning from query construction and allows sample acquisition to exploit the geometry of the representation space rather than relying exclusively on classifier confidence.

2.1. Randomized-network representation module

The representation module maps each unlabeled RGB image into a compact spatio-spectral embedding that supports diversity-aware query selection. Rather than relying on end-to-end feature extractors optimized by backpropagation, it uses randomized neural networks and randomized autoencoders with closed-form output-layer estimation. In the present work, $\Phi(\cdot)$ is instantiated through a spatio-spectral construction derived from ?, combining a spatial branch and a spectral branch.

Let I be an RGB image, and let I_C denote channel $C \in \{R, G, B\}$. For each channel, local 3×3 neighborhoods are extracted around every valid pixel position. Denoting the k -th neighborhood from channel C by $P_C^{(k)} \in \mathbb{R}^{3 \times 3}$, its vectorized form is

$$\mathbf{x}_C^{(k)} = \text{vec}\left(P_C^{(k)}\right) \in \mathbb{R}^9. \quad (4)$$

By stacking all vectorized neighborhoods column-wise, a feature matrix is obtained for each channel:

$$X_C = \left[\mathbf{x}_C^{(1)}, \mathbf{x}_C^{(2)}, \dots, \mathbf{x}_C^{(N)} \right], \quad (5)$$

where N is the number of extracted neighborhoods. Thus, each column of X_C corresponds to one local patch, while the full matrix summarizes the local spatial organization observed throughout channel C .

Given an input matrix $X \in \mathbb{R}^{p \times N}$ and an output matrix $Y \in \mathbb{R}^{r \times N}$, the randomized-network learning scheme first augments the input with a bias

term, producing X' , and computes the hidden representation

$$Z = \phi(WX'), \quad (6)$$

where $W \in \mathbb{R}^{Q \times (p+1)}$ is a randomly generated hidden-layer weight matrix, $\phi(\cdot)$ is a nonlinear activation function, and Q is the number of hidden neurons. The output-layer weights are then estimated in closed form by

$$M = YZ^T(ZZ^T + \lambda I)^{-1}, \quad (7)$$

where $\lambda > 0$ is a Tikhonov regularization parameter.

In the spatial branch (Fig. 1), a channel-wise matrix X_C is used both as input and reconstruction target, yielding

$$Z_C = \phi(WX'_C), \quad M_C = X_C Z_C^T (Z_C Z_C^T + \lambda I)^{-1}, \quad (8)$$

from which the spatial signature is obtained as

$$\Theta_C(Q) = \text{stretch}(M_C). \quad (9)$$

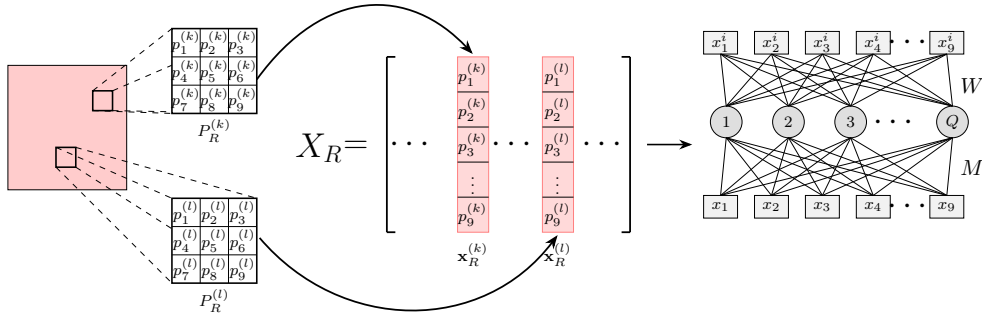


Figure 1: Illustration of the spatial component of the randomized-network representation module used in RNHAL (inspired by ?). Each local 3×3 neighborhood extracted from the red channel is vectorized into a column $\mathbf{x}_R^{(k)} = \text{vec}(P_R^{(k)})$, and the resulting matrix X_R is used to estimate the spatial signature $\Theta_R(Q)$ through a randomized autoencoder with closed-form output-layer learning.

In the spectral branch (Fig. 2), the matrix of one channel is used as input and the matrix of an adjacent channel is used as target output. For channels

65 C_1 and C_2 ,

$$Z_{C_1} = \phi(WX'_{C_1}), \quad M_{C_1, C_2} = X_{C_2}Z_{C_1}^T(Z_{C_1}Z_{C_1}^T + \lambda I)^{-1}, \quad (10)$$

66 and the corresponding spectral signature is

$$\Omega_{C_1, C_2}(Q) = \text{stretch}(M_{C_1, C_2}). \quad (11)$$

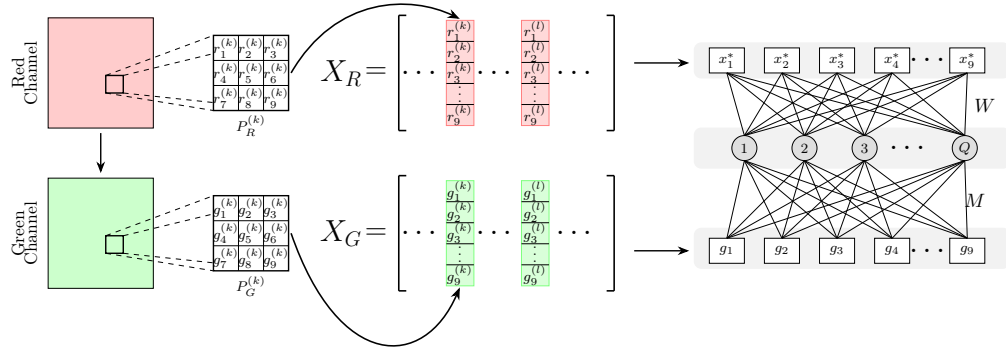


Figure 2: Illustration of the spectral component of the randomized-network representation module used in RNHAL. Corresponding local 3×3 neighborhoods are extracted from the red and green channels and vectorized into the matrices X_R and X_G , respectively. A randomized network is then trained to predict X_G from X_R , and the resulting output-layer weights are stretched to form the spectral signature $\Omega_{R,G}(Q)$.

67 The spatial branch captures within-channel local organization, whereas
 68 the spectral branch captures predictive relations between adjacent channels
 69 such as (R, G) , (G, B) , and (B, R) . The final representation of image x is
 70 formed by concatenation:

$$\Phi(x) = [\Theta_R(Q), \Theta_G(Q), \Theta_B(Q), \Omega_{R,G}(Q), \Omega_{G,B}(Q), \Omega_{B,R}(Q)]. \quad (12)$$

71 Thus, the resulting embedding integrates intra-channel spatial structure
 72 and inter-channel spectral dependencies into a single representation. Within
 73 RNHAL, $\Phi(\cdot)$ is not used as the final predictive classifier; instead, it struc-
 74 tures the unlabeled pool for query selection. Because $\Phi(\cdot)$ is independent of
 75 f_{θ_t} , the embeddings can be computed once and reused throughout the active

76 learning process. For the current unlabeled pool,

$$\mathcal{Z}_U^{(t)} = \left\{ \mathbf{z}_x \mid x \in \mathcal{D}_U^{(t)} \right\}. \quad (13)$$

77 The hierarchical acquisition stage then operates on $\mathcal{Z}_U^{(t)}$ rather than directly
 78 on classifier confidence scores.

79 2.2. Hierarchical selection module

80 The second stage of RNHAL builds a hierarchical partition of the em-
 81 bedding set $\mathcal{Z}_U^{(t)}$ through successive clustering steps. Let $\mathcal{Z}_U^{(t)} = \{\mathbf{z}_x \mid x \in$
 82 $\mathcal{D}_U^{(t)}\} \subset \mathbb{R}^d$ denote the current unlabeled pool in the embedding space. The
 83 hierarchical construction takes as input the number of levels L , the number of
 84 clusters per level $(k_\ell)_{\ell=1}^L$, the number of resampling steps m , and the number
 85 of resampled points per cluster $(r_\ell)_{\ell=1}^L$.

86 At the first level, k-means is applied directly to $\mathcal{Z}_U^{(t)}$, producing k_1 clusters
 87 and their corresponding centroids. At each subsequent level $\ell \in \{2, \dots, L\}$,
 88 the input is given by the centroids obtained at level $\ell - 1$, which are clustered
 89 into k_ℓ higher-level groups. Repeating this procedure recursively produces a
 90 tree-structured hierarchy in which lower levels capture fine-grained local vari-
 91 ation and higher levels summarize broader regions of the embedding space.

92 To reduce the tendency of flat clustering to over-fragment dense regions,
 93 each level is further refined through a resampling-clustering procedure. After
 94 the initial clustering at level ℓ , the algorithm performs m resampling steps. In
 95 each step, r_ℓ points closest to the centroid of each cluster are selected, and the
 96 union of these points forms a resampled set. A new k-means is then computed
 97 on this resampled set, generating an updated set of centroids for level ℓ .
 98 Finally, all input points of that level are reassigned to the updated centroids.
 99 This resampling-clustering cycle is repeated m times before proceeding to
 100 the next hierarchical level. The overall process is illustrated in Fig. 3.

101 The hierarchical organization provides a more stable and less density-
 102 dominated summary of the unlabeled pool, helping to avoid a disproportion-
 103 ate allocation of clusters to dominant regions of the representation space. As
 104 a result, the final hierarchy $\mathcal{T}_t$ supports a more balanced organization of the
 105 unlabeled data across multiple levels of granularity.

106 Once the hierarchy $\mathcal{T}_t$ has been constructed, the query budget n_{query} is
 107 allocated in a top-down manner. Starting from the highest level, the budget
 108 assigned to a parent node is distributed among its children, and the same

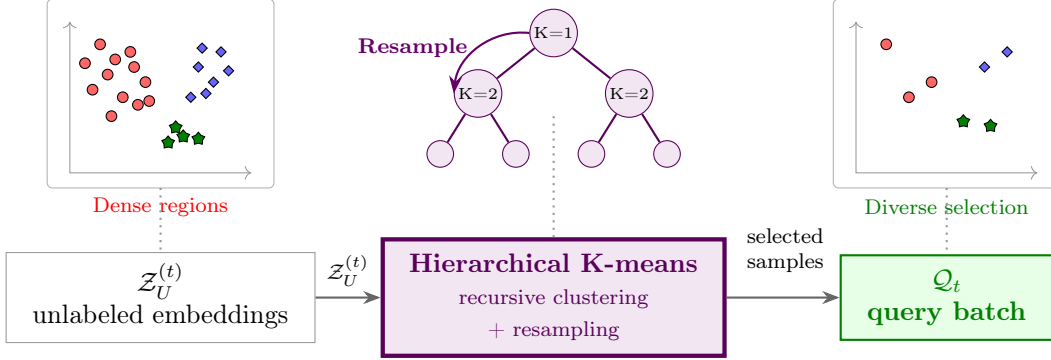


Figure 3: Hierarchical selection module of RNHAL. The embedding pool $\mathcal{Z}_U^{(t)}$ is recursively partitioned through hierarchical k-means with resampling, producing a multi-level organization of the unlabeled data. Query samples are then selected across the hierarchy to improve coverage and intra-batch diversity while reducing concentration in dense regions of the representation space.

109 allocation procedure is applied recursively until the leaf clusters are reached.
 110 The final query batch is then obtained by sampling instances from the se-
 111 lected leaf clusters. Therefore, the queried set is defined as

$$\mathcal{Q}_t = \Gamma_{n_{\text{query}}} \left(\mathcal{D}_U^{(t)}; g_t^{\text{RNHAL}} \right), \quad (14)$$

112 where g_t^{RNHAL} is induced by the representation function $\Phi(\cdot)$ and the hierar-
 113 chy $\mathcal{T}_t$ built over $\mathcal{Z}_U^{(t)}$. In this way, the hierarchical module does not merely
 114 cluster the embeddings, but it transforms the representation space into a
 115 coverage-aware query policy for batch active learning.

116 *Algorithmic description.* Algorithm 4 summarizes the proposed RNHAL pro-
 117 cedure. Since the randomized-network representation module is independent
 118 of the predictive classifier trained during active learning, the embedding of
 119 each sample is computed only once. Thus, before the iterative active learning
 120 rounds begin, all images in the data pool are mapped into the representa-
 121 tion space by $\Phi(\cdot)$. Then, at each round, the hierarchical selection module is
 122 applied only to the embeddings associated with the current unlabeled pool,
 123 from which a query batch of size n_{query} is selected. The queried instances
 124 are sent to annotation, incorporated into the labeled set, and the predictive
 125 classifier is retrained in the next round.

Require: Full data pool $\mathcal{D}$, initial labeled set $\mathcal{D}_L^{(0)}$, initial unlabeled set $\mathcal{D}_U^{(0)}$, query size n_{query} , number of rounds T

Ensure: Queried batches $\{\mathcal{Q}_t\}_{t=0}^{T-1}$

- 1: Precompute $\mathbf{z}_x = \Phi(x)$ for all $x \in \mathcal{D}$
- 2: **for** $t = 0$ to $T - 1$ **do**
- 3: Train predictive classifier f_{θ_t} on $\mathcal{D}_L^{(t)}$
- 4: $\mathcal{Z}_U^{(t)} \leftarrow \{\mathbf{z}_x \mid x \in \mathcal{D}_U^{(t)}\}$
- 5: $\mathcal{T}_t \leftarrow \text{HierarchicalKMeans}(\mathcal{Z}_U^{(t)})$
- 6: $\mathcal{Q}_t \leftarrow \text{BalancedSample}(\mathcal{T}_t, n_{\text{query}})$
- 7: Annotate samples in $\mathcal{Q}_t$
- 8: $\mathcal{D}_L^{(t+1)} \leftarrow \mathcal{D}_L^{(t)} \cup \mathcal{Q}_t$
- 9: $\mathcal{D}_U^{(t+1)} \leftarrow \mathcal{D}_U^{(t)} \setminus \mathcal{Q}_t$
- 10: **end for**

Figure 4: RNHAL procedure. The randomized-network embeddings are computed once for the full data pool, whereas hierarchical query selection and predictive model retraining are updated at each active learning round.

3. Experimental Setup

3.1. Dataset

The dataset used in this study consists of UAV-acquired RGB weed images extracted from georeferenced orthomosaics collected over agricultural fields in Brazil. The surveyed areas comprised 282 farms located in the states of São Paulo and Mato Grosso do Sul. Due to contractual privacy agreements, the exact geographic coordinates are not disclosed. The imagery was acquired across multiple field campaigns conducted at different times of the year, under heterogeneous operational conditions. To preserve the realism of the scenario, no explicit control was imposed on illumination conditions, crop or weed phenological stage, or infestation density. The resulting imagery has a ground sampling distance (GSD) of approximately 3 cm per pixel.

The orthomosaics were partitioned into non-overlapping patches of 128×128 pixels, and only patches containing weeds were retained for annotation. Each patch was assigned a single image-level label corresponding to one weed category. In total, the dataset contains 10,192 image patches divided into two disjoint subsets: an unlabeled pool with 8,086 samples and a fixed test

144 set with 2,106 samples. Five weed categories were considered: *Gramineae*
 145 spp., *Ricinus communis*, *Brachiaria* spp., *Panicum maximum*, and broadleaf
 146 species. Representative examples are shown in Fig. 5.

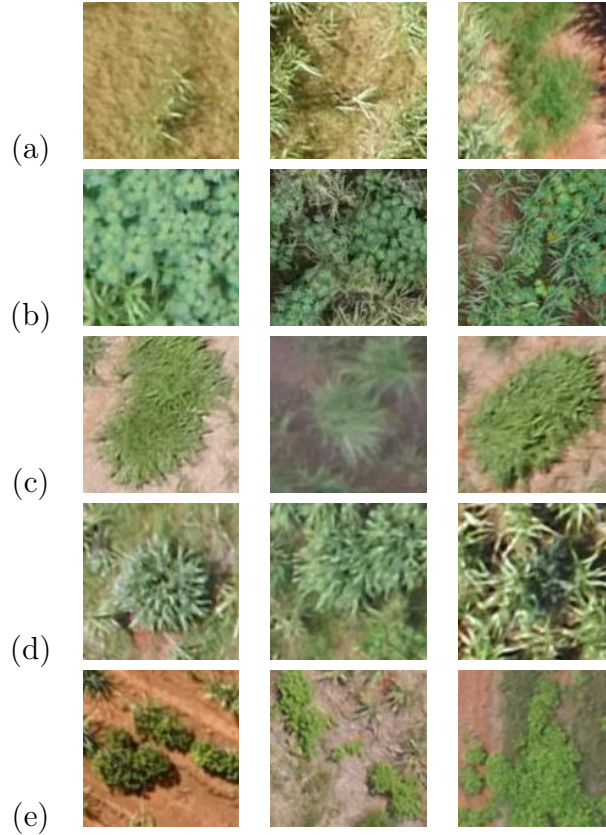


Figure 5: Representative sample images of each class in the dataset under realistic environmental conditions: (a) *Gramineae* spp., (b) *Ricinus communis*, (c) *Brachiaria* spp., (d) *Panicum maximum*, and (e) broadleaf species.

147 Image annotation was performed by a human specialist with agricultural
 148 domain expertise, who manually assigned each patch to its corresponding
 149 class. In the few cases where visual interpretation remained uncertain, field
 150 campaigns were conducted to validate the correct label. During the active
 151 learning experiments, the pool subset was treated as unlabeled only from
 152 the algorithmic point of view, although all pool samples were pre-annotated
 153 offline to enable a consistent and reproducible simulation of the annotation
 154 cycle. The test subset remained fixed throughout all experiments.

155 The class distribution is imbalanced, which is consistent with practical weed monitoring scenarios. In the pool set, the class counts are 1,198
 156 (*Gramineae* spp.), 3,370 (*Ricinus communis*), 774 (*Brachiaria* spp.), 1,036
 157 (*Panicum maximum*), and 1,708 (broadleaf species). In the fixed test set, the
 158 corresponding counts are 970, 456, 335, 190, and 155, respectively, as summarized in Table 1. All images were normalized using the standard ImageNet
 159 statistics to ensure compatibility with ImageNet-pretrained backbones.
 160
 161

Dataset class	Size Pool set	Size Test set
Gramineae spp.	1,198	970
Ricinus communis	3,370	456
Brachiaria spp.	774	335
Panicum maximum	1,036	190
Broadleaf species	1,708	155
Total	8,086	2,106

Table 1: Class distribution in the unlabeled pool and fixed test set.

162 3.2. Training protocol

163 All experiments were conducted within a pool-based active learning setting. At the beginning of each run, an initial labeled set of 100 images was
 164 randomly selected from the unlabeled pool, while the remaining pool samples
 165 were treated as candidate instances for query selection. At each active learning
 166 round, a predictive classifier f_{θ_t} was trained on the currently labeled set,
 167 the query mechanism selected a batch of unlabeled samples for annotation,
 168 and the newly labeled instances were incorporated into the training set for
 169 the next round. The test set remained fixed throughout all experiments.
 170

171 The predictive backbone adopted in all experiments was ResNet-50, initialized with ImageNet-pretrained weights and fine-tuned for the weed classification task. This architecture was chosen because of its strong generalization ability and its widespread use as a reliable baseline in image classification problems. At each active learning round, the network was retrained for 10
 172 epochs using the updated labeled subset and evaluated on the fixed test set,
 173 allowing a controlled comparison of the different acquisition strategies under
 174 the same classifier and training conditions.
 175

176 To evaluate the effect of annotation budget on active learning performance, three query sizes were considered: $n_{\text{query}} \in \{10, 50, 100\}$. Each experiment comprised eight active learning rounds. Therefore, the final number
 177
 178
 179
 180
 181

of labeled samples depended on the query size. Starting from the 100 initially labeled images, the final labeled sets contained 180, 500, and 900 samples for query sizes of 10, 50, and 100, respectively. In all cases, the number of labeled samples remained substantially smaller than the full pool of 8,086 images.

To ensure robustness, each experimental configuration was repeated three times using different random seeds. Considering the three query sizes and the 14 evaluated acquisition strategies, a total of 126 active learning runs were performed. In addition to these experiments, an upper-bound reference was obtained by training the same classifier using the full pool of labeled images. This reference provides an estimate of the maximum predictive performance achievable when no annotation budget constraint is imposed.

The compared acquisition strategies included RandomSampling, LeastConfidence, MarginSampling, EntropySampling, LeastConfidenceDropout, MarginSamplingDropout, EntropySamplingDropout, BALDDropout, KMeansSampling, KCenterGreedy, AdversarialBIM, AdversarialDeepFool, and the two proposed hierarchical methods based on spatio-spectral randomized representations. Random sampling was included as a baseline in all experiments. All strategies were evaluated under the same experimental protocol, enabling a standardized comparison between uncertainty-based, diversity-based, adversarial, and representation-guided query mechanisms.

3.3. Evaluation metrics

Model performance was evaluated using metrics derived from the confusion matrix, namely true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). Since this is a multiclass classification problem, the metrics were computed for each class and then aggregated using a weighted average in order to account for class imbalance.

Precision, recall, and F1-score were computed as

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (15)$$

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (16)$$

and

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (17)$$

Due to the significant class imbalance in the dataset, the F1-score was adopted as the primary evaluation metric. In this setting, overall accuracy can be misleading because it may be dominated by the performance of the most frequent classes, whereas the F1-score provides a more balanced assessment by jointly accounting for precision and recall. In addition, confusion matrices were used to analyze class-specific error patterns and support the qualitative interpretation of the results.

4. Results and Discussion

This section evaluates the proposed RNHAL framework under different annotation budgets and compares its behavior against uncertainty-based, diversity-based, and adversarial active learning baselines. We first examine the learning dynamics across query budgets, emphasizing convergence and label efficiency. We then summarize the final quantitative results after eight active learning rounds and discuss their implications for label-efficient weed image classification in realistic agricultural scenarios.

4.1. Ablation study

4.1.1. Representation ablation

4.1.2. Hierarchy ablation

4.1.3. Contribution of the two RNHAL stages

4.2. Comparison with other approaches

We begin by establishing the fully supervised upper bound obtained by training the predictive classifier on the complete pool of 8,086 labeled images. Under this setting, the ResNet-50 backbone achieved an F1-score of 0.9134 (± 0.0084), which serves as the reference for the maximum attainable performance on this dataset. The active learning analysis then evaluates how closely different query strategies approach this upper bound while using substantially fewer annotated samples.

Figures 6, 7, and 8 show the F1-score progression over eight active learning rounds for $n_{\text{query}} = 10, 50$, and 100, respectively. Across all budgets, increasing the query size produced more stable learning curves and narrowed the gap to the fully supervised reference. More importantly, RNHAL achieved the strongest overall performance in every setting, indicating that representation-guided querying combined with hierarchical selection improves both robustness and label efficiency.

For the most restrictive budget, with $n_{\text{query}} = 10$, most methods showed limited gains, reflecting the difficulty of extracting sufficiently informative batches from very small queries. In this regime, RandomSampling reached a final F1-score of 0.5746 (± 0.0471), whereas the strongest conventional baseline, BALDDropout, achieved 0.6293 (± 0.0429). RNHAL obtained the best result, with a final F1-score of 0.6713 (± 0.0428), suggesting that under severe annotation constraints its main advantage lies in reducing redundancy while preserving structural diversity in the selected batch.

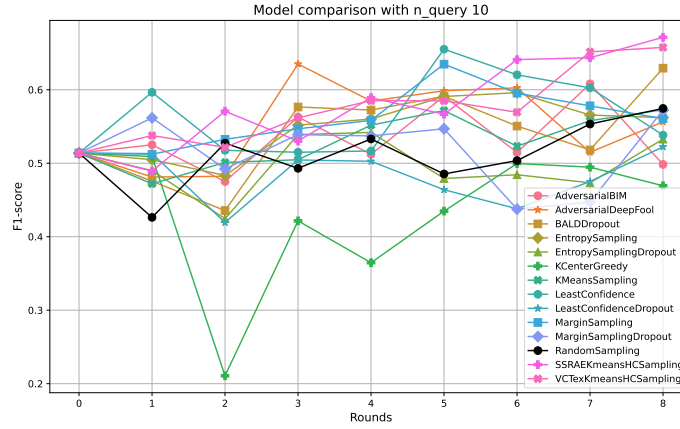


Figure 6: F1-score progression over 8 rounds for a query size of 10 samples per round.

With $n_{\text{query}} = 50$, the advantage of informed acquisition became substantially clearer. RandomSampling remained unstable and achieved a final F1-score of 0.5387 (± 0.2573), whereas several baselines benefited from the larger query budget. Among them, MarginSampling emerged as the strongest competitor, reaching 0.8023 (± 0.0272). RNHAL remained superior, with an F1-score of 0.8301 (± 0.0298). Notably, this result was obtained with only 500 labeled samples in total, including the initial labeled subset, showing that a moderate annotation budget can be translated into a highly informative training set when the query process is guided by representation quality and hierarchical coverage.

For the largest budget, with $n_{\text{query}} = 100$, the overall behavior became more stable across methods as the larger batches reduced the variance of the selection process. In this scenario, RandomSampling reached 0.8218 (± 0.0119), and the strongest conventional baseline, EntropySampling, achieved

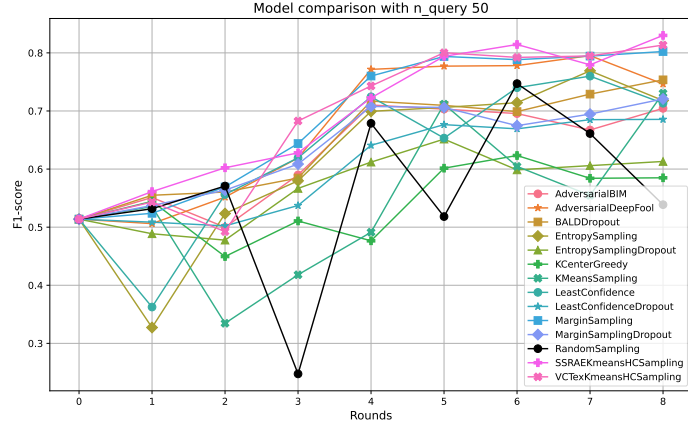


Figure 7: F1-score progression over 8 rounds for a query size of 50 samples per round.

0.8652 (± 0.0243). Even in this more favorable setting, RNHAL again produced the best overall result, with an F1-score of 0.8766 (± 0.0208). This performance was achieved with only 900 labeled samples, corresponding to approximately 11.13% of the full pool, while remaining close to the upper-bound F1-score of 0.9134.

Tables 2, 3, and 4 summarize the final classification results after eight active learning rounds in terms of accuracy, precision, recall, and F1-score. Because the dataset is imbalanced, the weighted F1-score is the main metric of interest. The tabulated results confirm the same pattern observed in the learning curves: under highly restrictive budgets, the gap between RNHAL and conventional baselines is already clear; under moderate and larger budgets, the proposed method remains consistently superior while all competitive methods move closer to the fully supervised reference.

For $n_{\text{query}} = 10$, the strongest non-proposed method was BALDDropout, with an F1-score of 0.6293 (± 0.0429), whereas RNHAL achieved 0.6713 (± 0.0428). For $n_{\text{query}} = 50$, the strongest baseline was MarginSampling, with 0.8023 (± 0.0272), while RNHAL reached 0.8301 (± 0.0298). For $n_{\text{query}} = 100$, EntropySampling provided the strongest baseline result, with 0.8652 (± 0.0243), but RNHAL again remained superior, achieving 0.8766 (± 0.0208). These comparisons indicate that the advantage of the proposed framework is not tied to a specific annotation budget, but persists from highly constrained to more permissive query regimes.

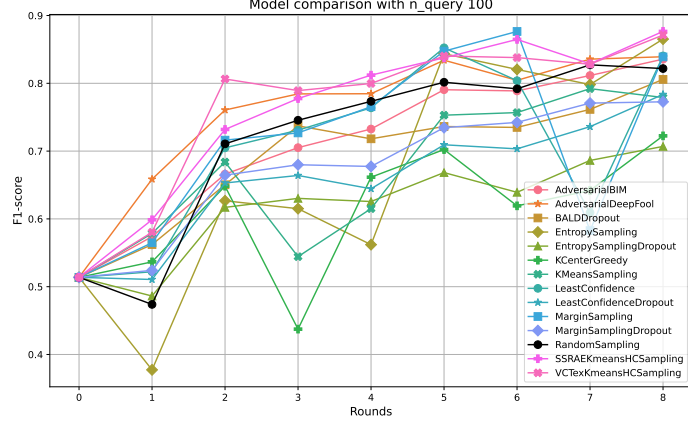


Figure 8: F1-score progression over 8 rounds for a query size of 100 samples per round.

Method	Accuracy	Precision	Recall	F1-score
AdversarialBIM	0.4789 (± 0.0460)	0.7362 (± 0.0534)	0.4789 (± 0.0460)	0.4984 (± 0.0518)
AdversarialDeepFool	0.5442 (± 0.0416)	0.7231 (± 0.0442)	0.5442 (± 0.0416)	0.5563 (± 0.0362)
BALDDropout	0.6223 (± 0.0466)	0.7152 (± 0.0174)	0.6223 (± 0.0466)	0.6293 (± 0.0429)
EntropySampling	0.5514 (± 0.0597)	0.7402 (± 0.0143)	0.5514 (± 0.0597)	0.5621 (± 0.0489)
EntropySamplingDropout	0.5207 (± 0.0878)	0.6912 (± 0.0354)	0.5207 (± 0.0878)	0.5328 (± 0.0908)
KCenterGreedy	0.4837 (± 0.0869)	0.6392 (± 0.0678)	0.4837 (± 0.0869)	0.4693 (± 0.1108)
KMeansSampling	0.5469 (± 0.0334)	0.7194 (± 0.0365)	0.5469 (± 0.0334)	0.5728 (± 0.0278)
LeastConfidence	0.5242 (± 0.0354)	0.7431 (± 0.0297)	0.5242 (± 0.0354)	0.5383 (± 0.0417)
LeastConfidenceDropout	0.5123 (± 0.1212)	0.7088 (± 0.0419)	0.5123 (± 0.1212)	0.5223 (± 0.1509)
MarginSampling	0.5519 (± 0.0772)	0.7283 (± 0.0281)	0.5519 (± 0.0772)	0.5606 (± 0.0867)
MarginSamplingDropout	0.5624 (± 0.0706)	0.7302 (± 0.0125)	0.5624 (± 0.0706)	0.5722 (± 0.0591)
RandomSampling	0.5562 (± 0.0549)	0.7065 (± 0.0054)	0.5562 (± 0.0549)	0.5746 (± 0.0471)
RNHAL	0.6648 (± 0.0520)	0.7433 (± 0.0231)	0.6648 (± 0.0520)	0.6713 (± 0.0428)

Table 2: Quantitative results for $n_query = 10$ after 8 active learning rounds (mean $\pm$ std).

Method	Accuracy	Precision	Recall	F1-score
AdversarialBIM	0.6808 (± 0.0428)	0.8162 (± 0.0336)	0.6808 (± 0.0428)	0.7045 (± 0.0406)
AdversarialDeepFool	0.7244 (± 0.0464)	0.8373 (± 0.0274)	0.7244 (± 0.0464)	0.7465 (± 0.0410)
BALDDropout	0.7387 (± 0.0171)	0.7986 (± 0.0350)	0.7387 (± 0.0171)	0.7536 (± 0.0241)
EntropySampling	0.6904 (± 0.0116)	0.8226 (± 0.0102)	0.6904 (± 0.0116)	0.7180 (± 0.0124)
EntropySamplingDropout	0.5935 (± 0.0371)	0.7247 (± 0.0457)	0.5935 (± 0.0371)	0.6131 (± 0.0330)
KCenterGreedy	0.5614 (± 0.1329)	0.6960 (± 0.1236)	0.5614 (± 0.1329)	0.5850 (± 0.1279)
KMeansSampling	0.7124 (± 0.0130)	0.8118 (± 0.0096)	0.7124 (± 0.0130)	0.7305 (± 0.0088)
LeastConfidence	0.6918 (± 0.0279)	0.8126 (± 0.0076)	0.6918 (± 0.0279)	0.7133 (± 0.0256)
LeastConfidenceDropout	0.6618 (± 0.0159)	0.7787 (± 0.0402)	0.6618 (± 0.0159)	0.6855 (± 0.0183)
MarginSampling	0.7889 (± 0.0312)	0.8566 (± 0.0134)	0.7889 (± 0.0312)	0.8023 (± 0.0272)
MarginSamplingDropout	0.7034 (± 0.0384)	0.7959 (± 0.0138)	0.7034 (± 0.0384)	0.7207 (± 0.0275)
RandomSampling	0.5632 (± 0.2045)	0.7148 (± 0.1172)	0.5632 (± 0.2045)	0.5387 (± 0.2573)
RNHAL	0.8210 (± 0.0354)	0.8577 (± 0.0154)	0.8210 (± 0.0354)	0.8301 (± 0.0298)

Table 3: Quantitative results for $n_query = 50$ after 8 active learning rounds (mean $\pm$ std).

Method	Accuracy	Precision	Recall	F1-score
AdversarialBIM	0.8200 (± 0.0216)	0.8858 (± 0.0120)	0.8200 (± 0.0216)	0.8356 (± 0.0192)
AdversarialDeepFool	0.8278 (± 0.0406)	0.8851 (± 0.0170)	0.8278 (± 0.0406)	0.8392 (± 0.0365)
BALDDropout	0.7942 (± 0.0168)	0.8455 (± 0.0022)	0.7942 (± 0.0168)	0.8057 (± 0.0125)
EntropySampling	0.8580 (± 0.0262)	0.8888 (± 0.0176)	0.8580 (± 0.0262)	0.8652 (± 0.0243)
EntropySamplingDropout	0.6853 (± 0.0118)	0.7906 (± 0.0434)	0.6853 (± 0.0118)	0.7069 (± 0.0141)
KCenterGreedy	0.7233 (± 0.1251)	0.7606 (± 0.1511)	0.7233 (± 0.1251)	0.7225 (± 0.1462)
KMeansSampling	0.7673 (± 0.0254)	0.8463 (± 0.0095)	0.7673 (± 0.0254)	0.7791 (± 0.0237)
LeastConfidence	0.8333 (± 0.0709)	0.8853 (± 0.0336)	0.8333 (± 0.0709)	0.8404 (± 0.0690)
LeastConfidenceDropout	0.7654 (± 0.0443)	0.8389 (± 0.0180)	0.7654 (± 0.0443)	0.7836 (± 0.0377)
MarginSampling	0.8243 (± 0.0395)	0.8894 (± 0.0155)	0.8243 (± 0.0395)	0.8390 (± 0.0344)
MarginSamplingDropout	0.7572 (± 0.0525)	0.8364 (± 0.0113)	0.7572 (± 0.0525)	0.7727 (± 0.0477)
RandomSampling	0.8150 (± 0.0106)	0.8570 (± 0.0107)	0.8150 (± 0.0106)	0.8218 (± 0.0119)
RNHAL	0.8688 (± 0.0237)	0.8983 (± 0.0119)	0.8688 (± 0.0237)	0.8766 (± 0.0208)

Table 4: Quantitative results for $n_query = 100$ after 8 active learning rounds (mean $\pm$ std).

290 A direct comparison with the fully supervised upper bound further high-
 291 lights the annotation efficiency of the proposed method. Training the clas-
 292 sifier on the entire pool yielded an F1-score of 0.9134 (± 0.0084), whereas
 293 RNHAL reached 0.8766 (± 0.0208) using only 900 labeled samples. This
 294 corresponds to an absolute reduction of 0.0368 in F1-score, while reduc-
 295 ing the volume of labeled data by 88.9%. In other words, the proposed
 296 representation-guided hierarchical selection strategy approaches fully super-
 297 vised performance with only a small fraction of the annotation cost.

298 These results also help explain the behavior observed in the learning
 299 curves. Standard deep active learning heuristics may exhibit unstable behav-
 300 ior in the earliest rounds, especially under restrictive query budgets, where
 301 predictive uncertainty is still poorly calibrated and random selection can ap-
 302 pear comparatively stable (?). By contrast, RNHAL remained consistently
 303 competitive across all evaluated budgets and achieved the best overall per-
 304 formance in every setting. This pattern suggests that its advantage is not
 305 based solely on informativeness in the uncertainty sense, but also on its abil-
 306 ity to reduce sampling redundancy and promote broader coverage of the
 307 unlabeled pool. Such behavior is consistent with recent findings showing
 308 that conventional uncertainty heuristics tend to over-sample visually similar
 309 majority instances in long-tailed settings, while under-representing minor-
 310 ity concepts and structurally diverse samples (?????). Overall, the results
 311 indicate that coupling randomized-network representations with hierarchi-
 312 cal batch selection provides a more label-efficient acquisition mechanism for
 313 visually heterogeneous agricultural data.

314 4.3. Class-wise and qualitative analysis

315 To complement the global metrics, we analyzed the class-specific behavior
 316 of the best-performing methods using confusion matrices obtained after eight
 317 active learning rounds with $n_{\text{query}} = 100$, which corresponds to the strongest
 318 overall experimental setting. In addition to RNHAL, we include the strongest
 319 conventional baselines and the fully supervised upper bound in order to assess
 320 how the proposed method affects the distribution of errors across classes.

321 The confusion matrices reveal that the main source of error is associ-
 322 ated with the three grass-like categories, namely *Gramineae* spp., *Brachiaria*
 323 spp., and *Panicum maximum*. This behavior is expected, since these classes
 324 exhibit strong visual similarity in canopy structure, leaf geometry, and tex-
 325 ture patterns, especially in UAV-derived orthomosaic patches with variable

illumination and background conditions. Even under these challenges, RN-HAL produced the cleanest class-wise behavior among the active learning strategies, indicating that the proposed representation-guided hierarchical acquisition process yields a more balanced and informative labeled subset.

The confusion matrix of RNHAL shows strong performance for the dominant class *Gramineae* spp., while also preserving good discrimination for *Brachiaria* spp., *Ricinus communis*, and broadleaf species. Although some confusion with *Panicum maximum* remains, the overall error pattern is more concentrated near visually adjacent classes, rather than dispersed across unrelated categories. This suggests that the proposed representation module captures relevant spatio-spectral structure, while the hierarchical selection mechanism improves class coverage and reduces the redundancy of the queried samples.

By contrast, the strongest uncertainty-based baselines still exhibit more pronounced class overlap. In particular, conventional heuristics such as EntropySampling and LeastConfidence show larger confusion between *Gramineae* spp. and *Panicum maximum*, as well as more unstable behavior for minority or morphologically ambiguous classes. This pattern is consistent with the quantitative results, which showed that these methods can remain competitive under larger query budgets, but still lag behind RNHAL in final F1-score. The comparison with the fully supervised upper bound further indicates that the remaining errors are largely tied to intrinsic visual ambiguity in the dataset rather than to clear failures of the proposed selection strategy.

To better interpret these class-wise results, we also inspected qualitative examples of correct and incorrect predictions produced by RNHAL under $n_{\text{query}} = 100$. Correct predictions show that the method is able to discriminate not only visually distinctive classes such as *Ricinus communis*, but also more difficult grass-like categories in the presence of soil interference, self-shadowing, and variable plant density. This indicates that the representation learned through the randomized-network module retains discriminative information that remains useful even under realistic field conditions.

The misclassified examples, on the other hand, reveal a consistent set of failure modes. The first corresponds to confusion among grass-like weeds with very similar texture and geometry, especially when plants are densely clustered or partially overlapping. The second involves confusion between *Ricinus communis* and broadleaf species at early phenological stages, when the characteristic lobed leaf structure of mature *Ricinus communis* is not yet clearly visible. These errors indicate that, although RNHAL substantially

364 improves label efficiency and class coverage, the task still contains biologically
365 ambiguous cases that are difficult to resolve from single RGB patches alone.

366 Overall, the class-wise and qualitative analyses support the same con-
367 clusion as the global metrics: the advantage of RNHAL lies not only in
368 improving the final F1-score, but also in producing a more balanced error
369 distribution across classes. In practical terms, this means that the proposed
370 method does not merely boost average performance; it also improves the reli-
371 ability of the training set acquired during active learning, which is especially
372 relevant in visually heterogeneous and long-tailed agricultural datasets.

373 4.4. Discussion

374 The results consistently show that RNHAL provides a more label-efficient
375 active learning strategy than conventional uncertainty-based, diversity-based,
376 and adversarial baselines. Across all evaluated query budgets, the proposed
377 method achieved the highest F1-score while requiring only a small fraction of
378 the full annotation cost. This behavior is particularly relevant in agricultural
379 applications, where expert annotation is expensive, visually ambiguous cases
380 are common, and class distributions are often strongly imbalanced.

381 A central finding of this study is that the benefit of RNHAL is not limited
382 to a single annotation regime. Under extremely restrictive budgets, its main
383 advantage lies in avoiding redundant acquisitions and promoting broader cov-
384 erage of the unlabeled pool. Under larger budgets, the same mechanism con-
385 tinues to provide gains, although the margin over strong baselines becomes
386 smaller as all methods benefit from increased supervision. This indicates
387 that the proposed framework is robust across different levels of annotation
388 availability, rather than being specialized for only one operating point.

389 From a methodological perspective, the results suggest that the superi-
390 ority of RNHAL is explained by the interaction between its two stages. The
391 randomized-network representation module provides a compact embedding
392 space that captures relevant spatio-spectral structure, while the hierarchi-
393 cal selection module organizes this space to allocate the query budget in a
394 way that favors both diversity and coverage. In this sense, the method does
395 not rely exclusively on predictive uncertainty. Instead, it uses the geome-
396 try of the representation space to guide batch construction, which appears
397 to be particularly beneficial in long-tailed and visually dense settings where
398 uncertainty-only heuristics often over-sample similar majority instances.

399 These observations are well aligned with the broader active learning liter-
400 ature. Standard uncertainty heuristics are known to become unstable in early

401 rounds, especially when the predictive model is poorly calibrated and the la-
402 beled set is still too small to support reliable confidence estimates. Likewise,
403 recent studies have shown that such methods can suffer from severe sampling
404 redundancy in long-tailed distributions, repeatedly querying samples from al-
405 ready dominant concepts while under-representing minority and structurally
406 diverse regions of the pool. The behavior observed here is consistent with
407 that view: although several conventional methods became competitive as
408 the query budget increased, RNHAL remained superior because it more ef-
409 fectively balanced informativeness with representational coverage.

410 From an application standpoint, these results highlight the relevance of
411 label-efficient learning for weed recognition in UAV imagery. In practical field
412 scenarios, reducing the number of required annotations without substantially
413 sacrificing predictive performance can directly lower the cost of building de-
414 ployable classification systems. In our experiments, RNHAL approached the
415 fully supervised upper bound while using only a small subset of the available
416 pool, indicating that strong performance can be achieved without exhaustive
417 labeling. This property is especially valuable for precision agriculture work-
418 flows, in which timely model updates and low annotation overhead are both
419 critical.

420 At the same time, the qualitative analysis makes clear that some er-
421 rors remain tied to the intrinsic difficulty of the data. In particular, con-
422 fusions among grass-like weeds and between early-stage *Ricinus communis*
423 and broadleaf species suggest that part of the remaining performance gap is
424 due to biological and visual ambiguity rather than to the acquisition strat-
425 egy alone. This points to a natural direction for future work: combining
426 label-efficient querying with richer sensing conditions, such as higher spatial
427 resolution, multispectral information, or multi-temporal imagery.

428 Taken together, the experimental evidence supports the main claim of
429 this study: RNHAL constitutes an effective active learning framework for
430 label-efficient weed image classification. By coupling randomized-network
431 representations with hierarchical batch selection, the method improves an-
432 notation efficiency, reduces sampling redundancy, and maintains strong per-
433 formance across a range of annotation budgets. These properties make it a
434 promising strategy for active learning in complex agricultural vision tasks.

435 *4.5. Discussion*

436 **5. Conclusion**

437 **Declaration of generative AI and AI-assisted technologies in the** 438 **manuscript preparation process**

439 The authors used generative artificial intelligence to assist in improving
440 the language and readability of this manuscript. The authors reviewed and
441 edited the text as necessary and take full responsibility for the content of
442 this publication.

443 **Data Availability**

444 For ethical, privacy, and security reasons related to the properties involved
445 in this study, the dataset used is not publicly available.

446 **References**